Short-Horizon Trajectory Forecasting from GPS-Derived Motion Features Using Deep Sequence Models

A Comparative Evaluation of GRU, LSTM, and TCN Models for Short-Term Motion Prediction

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ABSTRACT

This project evaluates short-horizon trajectory forecasting using only GPS-derived motion features from the Microsoft GeoLife dataset. The objective is to determine whether future position can be predicted one to five seconds ahead without relying on richer sensor inputs. To address this question, the study compares GRU, LSTM, and Temporal Convolutional Network models against a constant-velocity baseline using engineered kinematic features such as standardized positional deltas, speed, heading, acceleration, and turn rate.

The final pipeline uses trajectory-level data splitting, standardization, adaptive window limits, and streaming window generation to support stable training on long, heterogeneous GPS traces. Model performance is evaluated with displacement-based metrics in standardized coordinate space, allowing consistent comparison across architectures while avoiding interpretation of the results as physical distance errors.

Results show that all learned models outperform the constant-velocity baseline, with the tuned LSTM delivering the strongest overall displacement-error performance on validation and test data. The TCN becomes competitive after preprocessing and memory constraints are addressed, though it retains higher maximum errors than the LSTM. Overall, the findings show that accurate short-term motion forecasting is feasible with GPS-derived features alone and provide a practical foundation for future work in real-time mobility analytics.

KEYWORDS

Short-horizon trajectory forecasting; GPS-derived motion features; GeoLife dataset; deep learning; GRU; LSTM; Temporal Convolutional Network; kinematic feature engineering; mobility analytics; real-time prediction.

1 Introduction

This project addresses the problem of short-horizon trajectory forecasting using only timestamps and GPS coordinates, with the goal of predicting an agent's future position one to five seconds ahead. The central question guiding this work is: How accurately can we forecast near-term motion using only raw latitude, longitude, and time, without access to richer sensor suites such as IMUs, cameras, or wheel-speed sensors?

This problem is important because many operational systems must function in settings where sensor availability is limited, degraded, or inconsistent. A model capable of inferring short-term motion from sparse GPS feeds can support downstream tasks such as anomaly detection, collision risk estimation, and real-time decision support. In practical deployments, this capability can improve safety, reduce operational risk, and enhance situational awareness without requiring expensive or complex hardware.

The primary end users of this system depend on the deployment context. In operational environments, engineers and analysts may use trajectory forecasts to detect deviations from expected behavior or to support automated decision-making. In research or educational settings, the end user is the practitioner evaluating model performance and exploring the limits of GPS-only forecasting. In both cases, the value lies in producing accurate short-term predictions that could eventually be integrated into larger analytical or control pipelines.

This project uses the Microsoft GeoLife GPS Trajectory Dataset, a large-scale collection of real-world GPS trajectories recorded by 182 users over a five-year period. Each trajectory includes timestamped latitude, longitude, and altitude observations that support the derivation of motion features such as speed, heading, acceleration, and turn rate. Because many trajectories are sampled at short intervals and contain realistic GPS noise across diverse movement patterns, the dataset is well suited for evaluating GPS-only short-horizon forecasting.

The overarching goal of this project is to build a lightweight forecasting pipeline capable of predicting future positions over short horizons. The working hypothesis is that, even with limited inputs, derived kinematic features combined with sequence-based neural models such as LSTMs or temporal convolutional networks can achieve accurate and stable predictions. The final deliverable is an evaluated prototype pipeline that accepts sequences of GPS-derived motion features and outputs predicted positions for the next several seconds. Model performance is evaluated using displacement metrics, training stability, and visual comparisons between predicted and actual trajectories.

2 Data Summary

The GeoLife GPS Trajectory Dataset provides time-ordered GPS traces with timestamped latitude, longitude, and altitude observations. For this project, its value lies in its dense sampling, realistic GPS noise, and heterogeneous movement patterns, which together make it suitable for evaluating GPS-only short-horizon forecasting.

Preprocessing focused on converting raw GPS traces into stable motion sequences while avoiding unnecessary transformations. Rather than converting latitude and longitude into a local East-North-Up frame, the final pipeline operated in standardized latitude/longitude space. Standardization parameters were computed from the training trajectories, and each sliding window was additionally centered locally to improve numerical stability. Because of this design choice, displacement errors reported later are interpreted as standardized coordinate errors rather than physical distances in meters.

The main engineered features were standardized positional deltas, speed, heading, acceleration, and turn rate. Speed and acceleration were computed from position changes and elapsed time, while heading and turn rate were derived from bearing changes between consecutive GPS points. Explicit time-delta calculations were used because irregular sampling intervals can distort motion estimates. Heading wraparound at $\pm\pi$ was also identified as a source of possible turn-rate spikes and is noted as a future preprocessing improvement.

Exploratory Data Analysis

Exploratory analysis focused on whether the derived variables contained learnable short-term motion patterns and whether preprocessing artifacts could affect model training. Figure 1 summarizes timestamp spacing between consecutive GPS observations. Most usable windows contained short sampling intervals, which supports one-to-five-second forecasting, but occasional larger gaps appeared in the trajectories. These gaps are important because they can create artificial spikes in speed, acceleration, and turn rate even when the underlying movement is smooth.

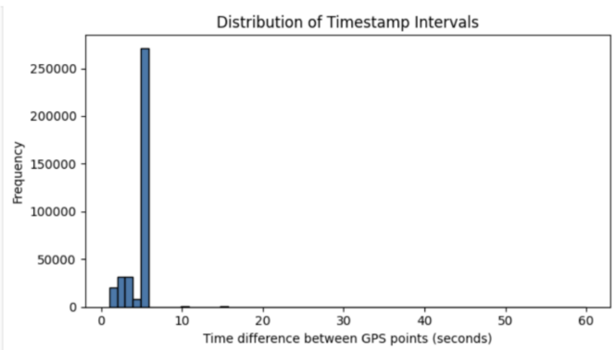

Figure 1. Distribution of timestamp intervals between consecutive GeoLife GPS observations. Most points occur at short intervals, while longer gaps identify irregular sampling that can introduce noise into derived motion features.

Figure 2 provides a representative speed profile for GeoLife trajectories. Speed profiles generally varied smoothly across adjacent timestamps, which supports using short sliding windows for near-term prediction. However, the speed plots also showed occasional spikes near irregular timestamp gaps, suggesting that some extreme values reflect sampling artifacts rather than true changes in movement. Acceleration was more volatile than speed, but its variation still aligned with recognizable movement behaviors such as gradual acceleration, steady cruising, and braking.

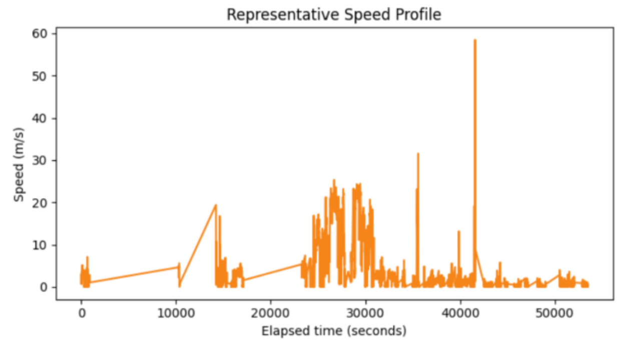

Figure 2. Representative speed profile from GeoLife trajectories. Speed generally changes smoothly over short horizons, with isolated spikes likely associated with timestamp irregularities or GPS noise.

Figure 3 shows the distributions of acceleration and turn rate, which provide additional evidence of temporal structure. Straight segments produced relatively stable headings and low turn rates, while turns produced distinct rotational changes. At the same time, abrupt heading wraparound created occasional artificial turn-rate spikes, reinforcing the need for careful angular preprocessing in future versions of the pipeline. Overall, the EDA suggests that the GeoLife data contain meaningful near-term motion dynamics while also requiring preprocessing to manage sampling irregularity and angular discontinuities.

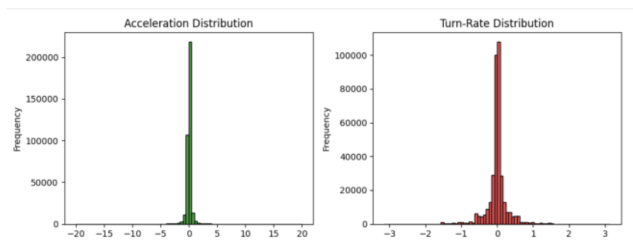

Figure 3. Distributions of acceleration and turn rate. Both variables are more volatile than speed and highlight the combined influence of real movement changes, irregular sampling, and heading discontinuities.

3 Literature Review

Short-horizon trajectory forecasting has been explored extensively across transportation analytics, mobility modeling, and geospatial prediction, with numerous methods designed to estimate near-term motion from positional data. Early approaches relied on classical kinematic models that assume constant velocity or constant acceleration. These models propagate future positions by integrating estimated speed and heading over short horizons and have been widely used in navigation and robotics applications due to their simplicity and low computational cost (Teunissen & Montenbruck, 2017). Although effective in stable motion regimes, these approaches struggle with nonlinear trajectories, sharp turns, and GPS noise, limiting their accuracy in real-world environments.

More sophisticated forecasting systems incorporate Bayesian filtering techniques such as the Kalman filter and its variants. These methods model motion as a dynamical system with latent velocity and acceleration states, enabling smoother estimates of position and heading and reducing the impact of GPS jitter. Kalman-based approaches have been applied extensively in vehicle tracking, pedestrian localization, and sensor fusion tasks, demonstrating improved stability over purely kinematic models (Sola, 2017). However, these methods often assume linear dynamics or require additional sensors such as IMUs to achieve high accuracy, making them less suitable for GPS-only scenarios.

Commercial mobility platforms also address short-term motion prediction using map-matching algorithms and route-constrained forecasting. Ride-sharing and navigation systems frequently combine GPS traces with road-network information to infer likely paths and estimate near-term movement (Zandbergen & Barbeau, 2011). Large-scale navigation systems such as Google Maps use learned motion models to predict driver arrival times and detect deviations from expected behavior, demonstrating the practical importance of short-horizon forecasting in operational environments (Derrow-Pinion et al., 2021).

Recent advances in machine learning have led to the adoption of sequence-based neural models for trajectory prediction. Recurrent neural networks, particularly long short-term memory (LSTM) networks and gated recurrent units (GRUs), were originally developed to capture long-range temporal dependencies in sequential data (Hochreiter & Schmidhuber, 1997; Cho et al., 2014). These architectures have since been adapted for mobility forecasting, where they learn nonlinear relationships between displacement, speed, heading, and future motion. Studies applying LSTMs and GRUs to pedestrian and vehicle trajectories have shown substantial improvements over classical baselines, especially in settings with irregular motion patterns or noisy sensor inputs (Zheng et al., 2008).

Temporal convolutional networks (TCNs) represent another major development in sequence modeling. By using dilated causal convolutions, TCNs can model long temporal contexts efficiently while maintaining stable gradients and low inference latency. Bai et al. (2018) demonstrated that TCNs outperform many recurrent architectures on sequence-prediction tasks, making them promising for real-time forecasting applications. Their ability to process sequences in parallel also makes them attractive for deployment in systems requiring rapid inference.

Several academic and commercial projects illustrate the application of these methods to problems similar to the one addressed in this capstone. LSTM-based models have been used to predict pedestrian trajectories in crowded environments, vehicle motion in urban traffic, and multimodal mobility patterns across large-scale GPS datasets (Zheng et al., 2008). GRU-based encoder–decoder architectures have been applied to short-term motion prediction in transportation networks, while TCNs have been used in real-time mobility analytics where low latency and temporal stability are critical (Bai et al., 2018). Commercial navigation systems employ variants of kinematic and learned motion models to estimate driver arrival times and detect anomalous movement. Collectively, these examples demonstrate that sequence-based neural architectures and derived kinematic features are well-suited for short-horizon trajectory forecasting using GPS-only inputs.

4 Methodology

The forecasting system developed for this project uses engineered kinematic features, sliding-window sequence extraction, and three deep learning architectures—GRU, LSTM, and a Temporal Convolutional Network (TCN)—to predict short-horizon motion from GPS-derived inputs. A classical constant-velocity baseline model is included to contextualize the performance of the learned models. This section describes the final model architecture, preprocessing pipeline, training procedures, and optimization strategy used in the project.

The forecasting problem is formulated as a supervised sequence-to-sequence task. Each model receives a fixed-length input window of twenty consecutive motion states and predicts the two-dimensional displacement for the next five time steps. Each motion state includes six engineered features derived from raw GPS coordinates: standardized latitude and longitude deltas, instantaneous speed, heading, linear acceleration, and turn rate. These variables capture both translational and rotational motion dynamics and are commonly used in mobility modeling and navigation systems. Although many GPS-based forecasting pipelines convert coordinates into a Cartesian frame such as ENU, the final implementation for this project operated entirely in standardized latitude/longitude space. All displacement metrics therefore reflect standardized coordinate units rather than physical distances in meters.

A simple kinematic baseline model was implemented to provide a lower-bound reference for evaluating the learned architectures. The baseline assumes constant velocity over the prediction horizon by extrapolating the most recent observed displacement. It computes the difference between the final two positions in the input window and repeats this displacement for each of the five future timesteps. Using the same window generator and evaluation metrics as the learned models ensures a fair comparison.

Three neural architectures were developed to explore different approaches to temporal modeling. The GRU model consists of a single gated recurrent unit layer with 64 hidden units, followed by a fully connected layer that maps the final hidden state to the ten displacement outputs. The LSTM model mirrors this structure, replacing the GRU layer with a long short-term memory layer of the same width. Both architectures are designed to capture temporal dependencies through gating mechanisms that regulate information flow.

The TCN model incorporates a deeper and more expressive architecture. It uses five temporal convolutional blocks with dilation factors of 1, 2, 4, 8, and 16, enabling the receptive field to span and exceed the twenty-step input window. Each block includes a dilated convolution, batch normalization, ReLU activation, dropout, and a residual connection. Sinusoidal positional encodings are added to the input sequence to provide temporal context, and a multi-head self-attention layer with four heads processes the temporal features after the convolutional stack. A final fully connected layer maps the attended representation to the displacement outputs. This architecture was selected to evaluate whether convolutional temporal modeling and attention mechanisms can outperform recurrent baselines in short-horizon forecasting.

Training follows a unified pipeline across all models. The GeoLife dataset is split at the trajectory level into 60 percent training, 20 percent validation, and 20 percent test sets to prevent leakage between segments of the same trajectory. Standardization parameters are computed from the training trajectories, and each sliding window is additionally centered locally to improve numerical stability. A streaming window generator produces batches of input windows and future displacement targets without loading all windows into memory simultaneously. Because some GeoLife trajectories contain tens of thousands of samples, adaptive caps are applied to limit the number of windows extracted from any single trajectory, ensuring consistent batch sizes and preventing memory exhaustion.

All models are trained using the Adam optimizer with a learning rate of 1×10^{-3} and mean squared error loss (Kingma & Ba, 2015). Training proceeds for twenty epochs, with progress and system-resource heartbeats logged periodically. Effective batch composition depends on the number of windows generated per trajectory, which varies with trajectory length and the adaptive caps. This approach maintains computational feasibility while preserving the temporal diversity of the dataset.

Model optimization involved iterative adjustments to architectural and training hyperparameters. Hidden sizes of 32, 64, and 128 were evaluated for the recurrent models, along with variations in dropout rates and learning rates. For the TCN, the number of convolutional layers, dilation schedule, and attention-head count were explored. Window caps were tuned to balance memory constraints with training diversity. These optimization experiments informed the final architecture selections described above; comparative performance outcomes are presented in the Results section.

5 Results

Model performance was evaluated using three displacement-based metrics commonly used in short-horizon trajectory forecasting: Average Displacement Error (ADE), Maximum Displacement Error (MDE), and Final Displacement Error (FDE). These metrics quantify how far predicted positions deviate from ground-truth future positions over the five-step prediction horizon. All metrics were computed directly in standardized latitude and longitude coordinate space. In this project, standardized coordinate units refer to coordinate values scaled using training-set statistics rather than min-max scaling to a 0–1 range. As a result, the reported displacement errors should be interpreted as errors in standardized coordinate units, not as meters, degrees, or proportions of a unit interval. Evaluation was conducted on both validation and held-out test trajectories to assess generalization.

The initial, pre-tuning results revealed clear differences in model stability and accuracy. The constant-velocity baseline performed poorly on MDE, reaching 8.448 on the test set in standardized coordinate units. This value does not represent 8.448 meters or a proportion on a 0–1 scale; instead, it indicates a large error relative to the training-set coordinate-displacement scale. This confirmed that simple kinematic extrapolation is insufficient for noisy GPS trajectories. The GRU and LSTM models substantially outperformed the baseline, achieving validation ADE values of 0.012 and 0.006, and test ADE values of 0.014 and 0.009. The LSTM consistently produced the lowest displacement errors among the recurrent models, which aligns with prior work showing its strength in sequential prediction tasks. The TCN model showed mixed behavior in the pre-tuning phase. Its validation ADE of 0.014 was competitive with the GRU, but its validation MDE of 0.765 and test ADE of 0.048 were significantly higher. This instability was traced to extremely long trajectories that produced tens of thousands of windows, which caused memory pressure and degraded training quality.

The tuned TCN achieved strong overall performance once memory constraints were addressed. Its validation ADE of approximately 0.0024 and test ADE of approximately 0.0025 were competitive with the recurrent models, while its MDE values remained higher than those of the LSTM. Although the TCN did not outperform the LSTM across the full set of standardized displacement metrics, its stability improved substantially compared to the pre-tuning phase, demonstrating the importance of preprocessing and window-generation optimizations.

As shown in Figure 4, all three models exhibit smooth and monotonic decreases in training loss, with no divergence between training and validation splits. This indicates that none of the tuned models exhibited severe overfitting. The addition of per-window centering and adaptive window caps was essential for preventing instability, particularly in the TCN. These optimizations directly addressed the irregular sampling intervals and long-trajectory artifacts present in the GeoLife dataset.

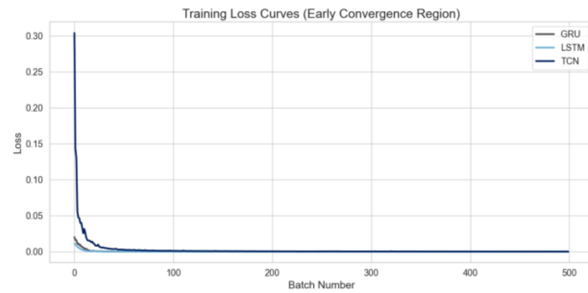

Figure 4. Training-loss curves for the GRU, LSTM, and TCN models. All three models show stable and monotonic convergence.

The displacement-error distributions for the learned models on the held-out test set are shown in Figure 5. The GRU, LSTM, and TCN models all exhibit sharply concentrated densities near zero, with the LSTM showing the tightest cluster. These distributions visually reinforce the quantitative results reported earlier, where the tuned LSTM achieved the lowest ADE and MDE values among the learned models.

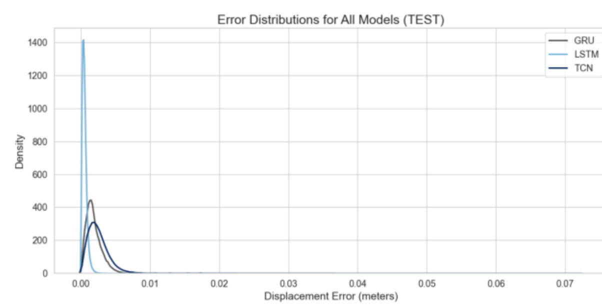

Figure 5. Kernel-density estimates of displacement-error distributions for the GRU, LSTM, and TCN models on the test set.

A visual comparison of predicted versus true final displacements is shown in Figure 6. Each point represents the predicted five-step displacement relative to the true final position. The learned models cluster tightly around the origin, demonstrating that the GRU, LSTM, and TCN all produce accurate short-term displacement estimates in standardized coordinate space.

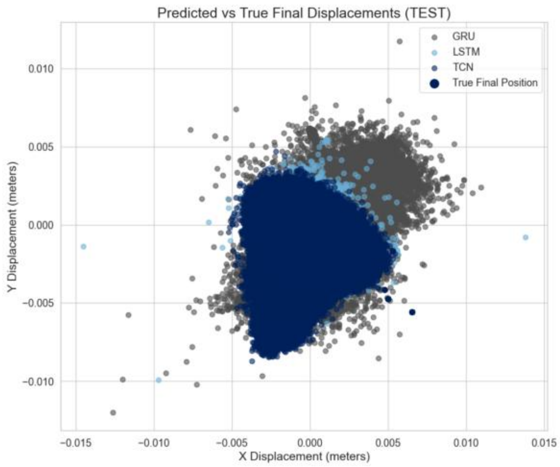

Figure 6. Scatter plot of predicted versus true final displacements for the GRU, LSTM, and TCN models on the test set.

After examining the displacement-error distributions and final-step scatter comparisons, a consolidated view of model performance across all displacement metrics is shown in Figure 7. This bar chart summarizes the ADE, FDE, and MDE values for the GRU, LSTM, and TCN models on the held-out test set. The LSTM achieves the lowest errors across all three metrics, reflecting its strong temporal stability and consistent short-horizon accuracy. The GRU performs slightly worse but remains competitive, while the TCN exhibits higher MDE values despite achieving low ADE and FDE. This visualization reinforces the earlier findings that the LSTM provides the most reliable overall performance, while the TCN's behavior is more sensitive to trajectory length and preprocessing conditions.

Figure 7. Comparison of ADE, FDE, and MDE values for the GRU, LSTM, and TCN models on the test set.

6 Conclusion

The goal of this project was to evaluate whether short-horizon motion can be accurately forecast using only GPS-derived features such as standardized positional deltas, speed, heading, acceleration, and turn rate, without relying on richer sensor suites. The working hypothesis was that sequence-based neural models would outperform a simple constant-velocity baseline and produce stable predictions relevant to future real-time mobility applications. The results strongly support this hypothesis. All learned models significantly outperformed the baseline, and the tuned LSTM achieved the strongest overall displacement-error performance in standardized coordinate space on both validation and test sets.

The project demonstrated that careful preprocessing plays a central role in stabilizing short-horizon forecasting. Adaptive window caps, per-window centering, and improved handling of long trajectories were essential for preventing instability, particularly in the TCN. These adjustments allowed the TCN to converge smoothly and achieve competitive performance alongside the recurrent models. Although the LSTM ultimately produced the lowest displacement errors across most metrics, the TCN showed strong potential once its architectural and memory requirements were properly managed.

The results also highlighted the sensitivity of short-horizon forecasting to irregular sampling intervals and angular discontinuities in heading. These artifacts introduced noise into derived kinematic variables and occasionally inflated displacement errors. Future work should incorporate heading unwrapping, timestamp interpolation, and more robust smoothing techniques to further stabilize the input features. Additional preprocessing improvements would likely enhance model performance, especially in trajectories with inconsistent sampling rates.

A valuable extension of this work would be to examine how different GPS sampling intervals affect model accuracy. The GeoLife dataset contains many trajectories sampled near one hertz, but real-world systems often operate at lower or variable rates due to device limitations, power constraints, or environmental conditions. Systematically degrading the sampling frequency and evaluating the resulting impact on ADE, FDE, and MDE would reveal the minimum viable sampling rate at which the learned models maintain reliable short-term forecasts. This analysis would provide practical guidance for deployment scenarios where GPS updates may arrive more slowly, and it would help determine whether additional temporal modeling or interpolation strategies are required to preserve accuracy under reduced sampling conditions.

Several additional extensions would strengthen the forecasting system and broaden its applicability. Transformer-based architectures represent a promising direction, as they can model long-range temporal dependencies and adapt to irregular sampling with appropriate standardization strategies. Evaluating performance across different transportation modes such as walking, biking, and driving would provide deeper insight into how the models behave under distinct motion regimes. Developing a production-ready inference module with real-time streaming, latency profiling, and monitoring would enable deployment in operational environments where rapid and reliable short-term motion prediction is required.

Overall, the results demonstrate that accurate short-horizon motion forecasting is feasible using only GPS-derived features and lightweight temporal models. The final pipeline provides a strong foundation for future work in real-time mobility analytics and highlights the importance of careful preprocessing, sampling-rate considerations, and architectural design when working with large and heterogeneous geospatial datasets.

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